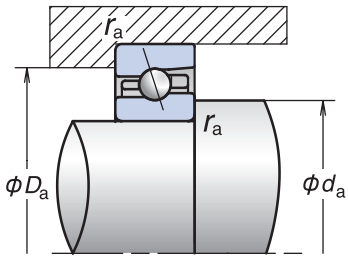
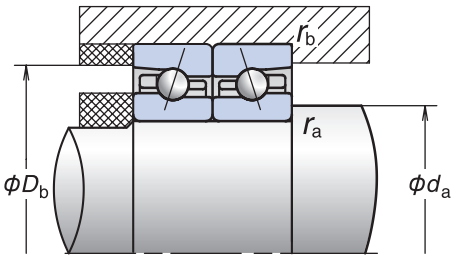


스핀 쇼트Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
55	67	69.5	0.6	0.3	49	95	240	499	40	51	78	111	0	− 4	− 13	− 24	0.130
55	67	69.5	0.6	0.3	49	100	266	565	44	58	89	129	0	− 4	− 13	− 24	0.114
55	67	69.5	0.6	0.3	71	154	379	791	94	124	175	235	− 1	− 4	− 10	− 18	0.132
55	67	69.5	0.6	0.3	74	168	430	911	107	143	204	276	− 1	− 4	− 10	− 18	0.116
55	67	69.5	0.6	0.3	49	162	319	–	44	68	89	–	0	− 8	− 16	–	0.127
55	67	69.5	0.6	0.3	49	177	359	–	49	78	103	–	0	− 8	− 16	–	0.111
55	67	69.5	0.6	0.3	49	177	359	–	49	78	103	–	0	− 8	− 16	–	0.111
55	67	69.5	0.6	0.3	49	253	511	–	73	128	166	–	0	− 8	− 15	–	0.127
55	67	69.5	0.6	0.3	49	283	584	–	81	149	194	–	0	− 8	− 15	–	0.111
55	67	69.5	0.6	0.3	49	283	584	–	81	149	194	–	0	− 8	− 15	–	0.111
55	67	69.5	0.6	0.3	49	162	319	–	44	68	89	–	0	− 8	− 16	–	0.144
55	67	69.5	0.6	0.3	49	177	359	–	49	78	103	–	0	− 8	− 16	–	0.128
55	67	69.5	0.6	0.3	49	177	359	–	49	78	103	–	0	− 8	− 16	–	0.128
55	67	69.5	0.6	0.3	49	253	511	–	73	128	166	–	0	− 8	− 15	–	0.144
55	67	69.5	0.6	0.3	49	283	584	–	81	149	194	–	0	− 8	− 15	–	0.128
55	67	69.5	0.6	0.3	49	283	584	–	81	149	194	–	0	− 8	− 15	–	0.128
56	74	75	1	0.5	70	152	388	791	46	63	95	135	− 2	− 8	− 20	− 34	0.259
56	74	75	1	0.5	73	165	438	906	51	72	110	158	− 2	− 8	− 20	− 34	0.228
56	74	75	1	0.5	125	262	580	1 189	118	154	208	278	− 3	− 7	− 14	− 24	0.270
56	74	75	1	0.5	136	293	664	1 379	135	179	244	327	− 3	− 7	− 14	− 24	0.239
56	74	75	1	0.5	49	243	525	1 299	113	194	256	360	0	− 5	− 10	− 20	0.262
56	74	75	1	0.5	49	119	250	–	47	64	86	–	0	− 5	− 12	–	0.281
56	74	75	1	0.5	49	128	280	–	52	74	100	–	0	− 5	− 12	–	0.266
56	74	75	1	0.5	49	128	280	–	52	74	100	–	0	− 5	− 12	–	0.266
56	74	75	1	0.5	49	270	552	–	78	140	182	–	0	− 8	− 15	–	0.281
56	74	75	1	0.5	49	303	633	–	87	162	213	–	0	− 8	− 15	–	0.266
56	74	75	1	0.5	49	303	633	–	87	162	213	–	0	− 8	− 15	–	0.266
56	74	75	1	0.5	49	119	250	–	47	64	86	–	0	− 5	− 12	–	0.330
56	74	75	1	0.5	49	128	280	–	52	74	100	–	0	− 5	− 12	–	0.315
56	74	75	1	0.5	49	128	280	–	52	74	100	–	0	− 5	− 12	–	0.315
56	74	75	1	0.5	49	270	552	–	78	140	182	–	0	− 8	− 15	–	0.330
56	74	75	1	0.5	49	303	633	–	87	162	213	–	0	− 8	− 15	–	0.315
56	74	75	1	0.5	49	303	633	–	87	162	213	–	0	− 8	− 15	–	0.315
57	83	85	1	0.6	127	248	590	1 171	52	69	102	143	− 7	− 15	− 31	− 50	0.457
57	83	85	1	0.6	137	277	674	1 351	60	80	119	167	− 7	− 15	− 31	− 50	0.387
57	83	85	1	0.6	208	391	989	1 934	130	164	233	306	− 6	− 11	− 23	− 37	0.453
57	83	85	1	0.6	231	445	1 145	2 257	150	191	274	361	− 6	− 11	− 23	− 37	0.383
57	83	85	1	0.6	49	477	1 156	2 038	105	227	314	390	0	− 10	− 20	− 30	0.462

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

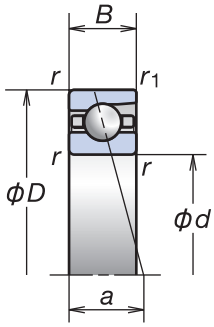
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

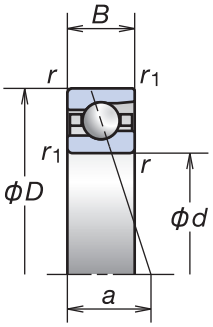
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

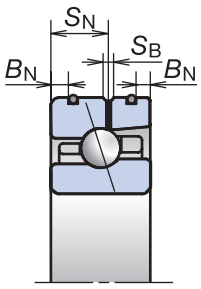
내경55mm



79,70,72계열
BNR10,20계열
BER10,20계열



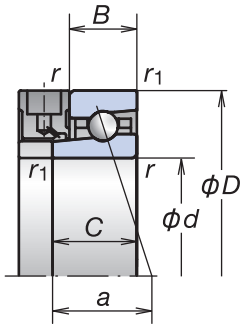
BNR19,29계열
BER19,29계열



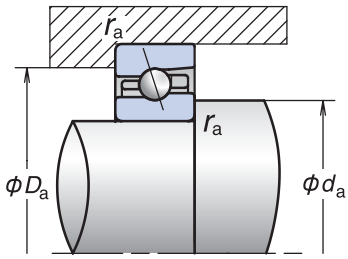
로바스트 쇼트

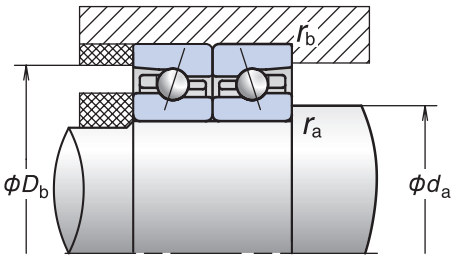
호칭 번호(1)	주요 치수(2) (mm)									기본 정격 하중(3) (kN)		허용(4) 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수(5) (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7911C	55	80	13	–	–	–	–	1	0.6	20.1	17.7	11.0	15	16.3	15.5	17 100	26 000
*7911CSN24	55	80	13	–	–	–	–	1	0.6	(20.1)	(17.7)	13.0	15	16.3	15.5	22 300	33 800
*7911A5	55	80	13	–	–	–	–	1	0.6	19.0	16.8	12.5	25	–	22.2	14 900	22 300
*7911A5SN24	55	80	13	–	–	–	–	1	0.6	(19.0)	(16.8)	14.8	25	–	22.2	19 300	28 900
*55BNR19E	55	80	13	–	–	–	–	1	0.6	14.4	11.4	16.2	18	10.7	17.5	22 300	31 800
*55BNR19H	55	80	13	–	2.8	7.5	1.4	1	0.6	(14.4)	(11.4)	10.6	18	10.7	17.5	26 700	41 500
*55BNR19X	55	80	13	18	2.8	7.5	1.4	1	0.6	(14.4)	(11.4)	10.6	18	10.7	17.5	31 200	48 900
*55BER19E	55	80	13	–	–	–	–	1	0.6	13.8	10.9	16.1	25	–	22.2	19 200	27 000
*55BER19H	55	80	13	–	2.8	7.5	1.4	1	0.6	(13.8)	(10.9)	12.9	25	–	22.2	23 800	37 100
*55BER19X	55	80	13	18	2.8	7.5	1.4	1	0.6	(13.8)	(10.9)	12.9	25	–	22.2	28 200	44 500
*55BNR29EV1V	55	80	16	–	–	–	–	1	0.6	14.4	11.4	16.2	18	10.7	19.0	22 300	–
*55BNR29HV1V	55	80	16	–	–	–	–	1	0.6	(14.4)	(11.4)	10.6	18	10.7	19.0	26 700	–
*55BNR29XV1V	55	80	16	–	–	–	–	1	0.6	(14.4)	(11.4)	10.6	18	10.7	19.0	31 200	–
*55BER29EV1V	55	80	16	–	–	–	–	1	0.6	13.8	10.9	16.1	25	–	23.7	19 200	–
*55BER29HV1V	55	80	16	–	–	–	–	1	0.6	(13.8)	(10.9)	12.9	25	–	23.7	23 800	–
*55BER29XV1V	55	80	16	–	–	–	–	1	0.6	(13.8)	(10.9)	12.9	25	–	23.7	28 200	–
*7011C	55	90	18	–	–	–	–	1.1	0.6	36.0	28.6	18.9	15	15.5	18.7	15 900	24 200
*7011CSN24	55	90	18	–	–	–	–	1.1	0.6	(36.0)	(28.6)	22.4	15	15.5	18.7	20 700	31 500
*7011A5	55	90	18	–	–	–	–	1.1	0.6	34.0	27.2	21.8	25	–	25.9	13 800	20 700
*7011A5SN24	55	90	18	–	–	–	–	1.1	0.6	(34.0)	(27.2)	25.8	25	–	25.9	18 000	26 900
*7011A	55	90	18	–	–	–	–	1.1	0.6	32.5	26.3	16.6	30	–	29.9	10 400	13 800
*55BNR10E	55	90	18	–	–	–	–	1.1	0.6	15.1	12.5	17.8	18	10.8	20.6	20 700	29 600
*55BNR10H	55	90	18	–	4.3	10.0	1.4	1.1	0.6	(15.1)	(12.5)	11.7	18	10.8	20.6	24 900	38 700
*55BNR10X	55	90	18	23	4.3	10.0	1.4	1.1	0.6	(15.1)	(12.5)	11.7	18	10.8	20.6	29 000	45 600
*55BER10E	55	90	18	–	–	–	–	1.1	0.6	14.4	12.0	21.0	25	–	25.7	17 800	25 200
*55BER10H	55	90	18	–	4.3	10.0	1.4	1.1	0.6	(14.4)	(12.0)	14.1	25	–	25.7	22 100	34 500
*55BER10X	55	90	18	23	4.3	10.0	1.4	1.1	0.6	(14.4)	(12.0)	14.1	25	–	25.7	26 300	41 400
*55BNR20EV1V	55	90	22	–	–	–	–	1.1	0.6	15.1	12.5	17.8	18	10.8	22.6	20 700	–
*55BNR20HV1V	55	90	22	–	–	–	–	1.1	0.6	(15.1)	(12.5)	11.7	18	10.8	22.6	24 900	–
*55BNR20XV1V	55	90	22	–	–	–	–	1.1	0.6	(15.1)	(12.5)	11.7	18	10.8	22.6	29 000	–
*55BER20EV1V	55	90	22	–	–	–	–	1.1	0.6	14.4	12.0	21.0	25	–	27.7	17 800	–
*55BER20HV1V	55	90	22	–	–	–	–	1.1	0.6	(14.4)	(12.0)	14.1	25	–	27.7	22 100	–
*55BER20XV1V	55	90	22	–	–	–	–	1.1	0.6	(14.4)	(12.0)	14.1	25	–	27.7	26 300	–
7211C	55	100	21	–	–	–	–	1.5	1	55.5	40.0	27.6	15	14.5	20.9	14 900	22 600
7211CSN24	55	100	21	–	–	–	–	1.5	1	(55.5)	(40.0)	32.8	15	14.5	20.9	19 400	29 500
7211A5	55	100	21	–	–	–	–	1.5	1	53.0	38.0	32.5	25	–	28.6	13 000	19 400
7211A5SN24	55	100	21	–	–	–	–	1.5	1	(53.0)	(38.0)	38.6	25	–	28.6	16 800	25 200
7211A	55	100	21	–	–	–	–	1.5	1	51.0	37.0	25.0	30	–	32.9	9 700	13 000

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다. 그리고 20, 29계열은 비접촉 씰사양만 대응 가능합니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



스핀 쇼트Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
61	74	75	1	0.5	60	111	296	593	45	58	90	126	− 1	− 5	− 15	− 26	0.182
61	74	75	1	0.5	61	119	331	675	50	66	104	147	− 1	− 5	− 15	− 26	0.163
61	74	75	1	0.5	99	196	454	917	113	144	198	264	− 2	− 5	− 11	− 19	0.184
61	74	75	1	0.5	106	217	517	1 059	129	167	232	309	− 2	− 5	− 11	− 19	0.165
61	74	75	1	0.5	49	167	333	–	46	72	94	–	0	− 8	− 16	–	0.178
61	74	75	1	0.5	49	184	375	–	51	82	109	–	0	− 8	− 16	–	0.158
61	74	75	1	0.5	49	184	375	–	51	82	109	–	0	− 8	− 16	–	0.158
61	74	75	1	0.5	49	264	538	–	76	136	176	–	0	− 8	− 15	–	0.178
61	74	75	1	0.5	49	296	615	–	85	158	206	–	0	− 8	− 15	–	0.158
61	74	75	1	0.5	49	296	615	–	85	158	206	–	0	− 8	− 15	–	0.158
61	74	75	1	0.5	49	167	333	–	46	72	94	–	0	− 8	− 16	–	0.213
61	74	75	1	0.5	49	184	375	–	51	82	109	–	0	− 8	− 16	–	0.194
61	74	75	1	0.5	49	184	375	–	51	82	109	–	0	− 8	− 16	–	0.194
61	74	75	1	0.5	49	264	538	–	76	136	176	–	0	− 8	− 15	–	0.213
61	74	75	1	0.5	49	296	615	–	85	158	206	–	0	− 8	− 15	–	0.194
61	74	75	1	0.5	49	296	615	–	85	158	206	–	0	− 8	− 15	–	0.194
62	83	85	1	0.6	95	200	479	971	51	69	102	144	− 4	− 11	− 24	− 40	0.380
62	83	85	1	0.6	100	221	544	1 118	58	80	119	169	− 4	− 11	− 24	− 40	0.332
62	83	85	1	0.6	157	345	804	1 552	129	171	236	307	− 4	− 9	− 18	− 29	0.383
62	83	85	1	0.6	172	391	928	1 807	148	199	277	362	− 4	− 9	− 18	− 29	0.335
62	83	85	1	0.6	49	246	887	1 307	115	196	311	360	0	− 5	− 15	− 20	0.385
62	83	85	1	0.6	49	121	305	–	49	67	95	–	0	− 5	− 14	–	0.414
62	83	85	1	0.6	49	131	343	–	54	77	111	–	0	− 5	− 14	–	0.393
62	83	85	1	0.6	49	131	343	–	54	77	111	–	0	− 5	− 14	–	0.393
62	83	85	1	0.6	49	356	725	–	81	160	208	–	0	− 10	− 18	–	0.414
62	83	85	1	0.6	49	404	836	–	90	186	244	–	0	− 10	− 18	–	0.393
62	83	85	1	0.6	49	404	836	–	90	186	244	–	0	− 10	− 18	–	0.393
62	83	85	1	0.6	49	121	305	–	49	67	95	–	0	− 5	− 14	–	0.501
62	83	85	1	0.6	49	131	343	–	54	77	111	–	0	− 5	− 14	–	0.480
62	83	85	1	0.6	49	131	343	–	54	77	111	–	0	− 5	− 14	–	0.480
62	83	85	1	0.6	49	356	725	–	81	160	208	–	0	− 10	− 18	–	0.501
62	83	85	1	0.6	49	404	836	–	90	186	244	–	0	− 10	− 18	–	0.480
62	83	85	1	0.6	49	404	836	–	90	186	244	–	0	− 10	− 18	–	0.480
64	91	94	1.5	0.8	142	289	788	1 554	56	74	116	163	− 8	− 17	− 38	− 60	0.601
64	91	94	1.5	0.8	155	323	906	1 802	64	86	136	191	− 8	− 17	− 38	− 60	0.502
64	91	94	1.5	0.8	248	491	1 206	2 397	142	182	256	338	− 7	− 13	− 26	− 42	0.596
64	91	94	1.5	0.8	277	562	1 401	2 804	165	213	301	399	− 7	− 13	− 26	− 42	0.497
64	91	94	1.5	0.8	49	494	1 722	2 635	108	236	372	438	0	− 10	− 26	− 35	0.609

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

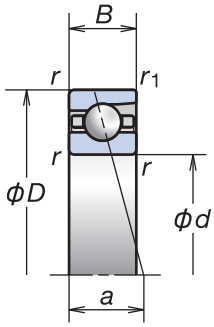
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

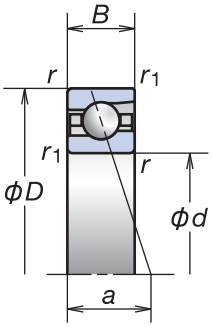
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

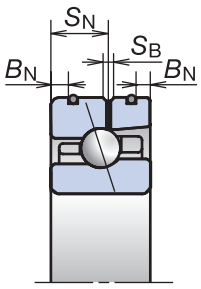
내경60mm



79,70,72계열
BNR10,20계열
BER10,20계열



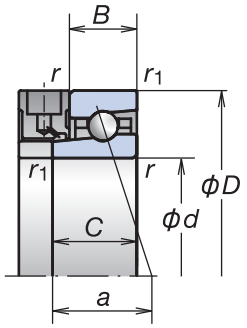
BNR19,29계열
BER19,29계열



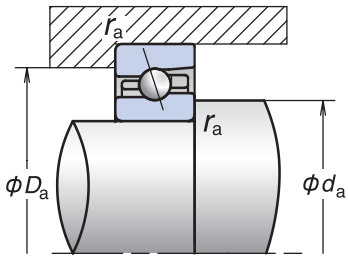
로바스트 쇼트

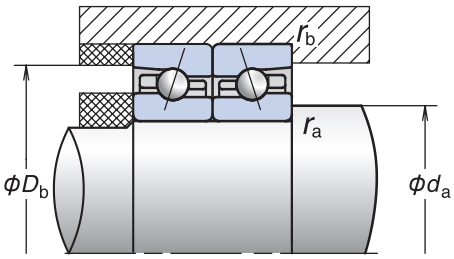
호칭 번호(1)	주요 치수(2) (mm)									기본 정격 하중(3) (kN)		허용(4) 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수(5) (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7912C	60	85	13	–	–	–	–	1	0.6	20.4	18.7	11.5	15	16.5	16.2	15 900	24 200
*7912CSN24	60	85	13	–	–	–	–	1	0.6	(20.4)	(18.7)	13.6	15	16.5	16.2	20 700	31 500
*7912A5	60	85	13	–	–	–	–	1	0.6	19.2	17.7	13.0	25	–	23.4	13 800	20 700
*7912A5SN24	60	85	13	–	–	–	–	1	0.6	(19.2)	(17.7)	15.5	25	–	23.4	18 000	26 900
*60BNR19E	60	85	13	–	–	–	–	1	0.6	14.6	12.0	17.1	18	10.7	18.3	20 700	29 600
*60BNR19H	60	85	13	–	2.8	7.5	1.4	1	0.6	(14.6)	(12.0)	11.2	18	10.7	18.3	24 900	38 700
*60BNR19X	60	85	13	18	2.8	7.5	1.4	1	0.6	(14.6)	(12.0)	11.2	18	10.7	18.3	29 000	45 600
*60BER19E	60	85	13	–	–	–	–	1	0.6	14.0	11.5	20.1	25	–	23.4	17 800	25 200
*60BER19H	60	85	13	–	2.8	7.5	1.4	1	0.6	(14.0)	(11.5)	13.6	25	–	23.4	22 100	34 500
*60BER19X	60	85	13	18	2.8	7.5	1.4	1	0.6	(14.0)	(11.5)	13.6	25	–	23.4	26 300	41 400
*60BNR29EV1V	60	85	16	–	–	–	–	1	0.6	14.6	12.0	17.1	18	10.7	19.8	20 700	–
*60BNR29HV1V	60	85	16	–	–	–	–	1	0.6	(14.6)	(12.0)	11.2	18	10.7	19.8	24 900	–
*60BNR29XV1V	60	85	16	–	–	–	–	1	0.6	(14.6)	(12.0)	11.2	18	10.7	19.8	29 000	–
*60BER29EV1V	60	85	16	–	–	–	–	1	0.6	14.0	11.5	20.1	25	–	24.9	17 800	–
*60BER29HV1V	60	85	16	–	–	–	–	1	0.6	(14.0)	(11.5)	13.6	25	–	24.9	22 100	–
*60BER29XV1V	60	85	16	–	–	–	–	1	0.6	(14.0)	(11.5)	13.6	25	–	24.9	26 300	–
*7012C	60	95	18	–	–	–	–	1.1	0.6	37.0	30.5	19.9	15	15.7	19.4	14 900	22 600
*7012CSN24	60	95	18	–	–	–	–	1.1	0.6	(37.0)	(30.5)	23.6	15	15.7	19.4	19 400	29 500
*7012A5	60	95	18	–	–	–	–	1.1	0.6	35.0	29.1	23.0	25	–	27.1	13 000	19 400
*7012A5SN24	60	95	18	–	–	–	–	1.1	0.6	(35.0)	(29.1)	27.3	25	–	27.1	16 800	25 200
*7012A	60	95	18	–	–	–	–	1.1	0.6	33.5	28.1	17.6	30	–	31.4	9 700	13 000
*60BNR10E	60	95	18	–	–	–	–	1.1	0.6	15.6	13.7	19.5	18	10.8	21.5	19 400	27 700
*60BNR10H	60	95	18	–	4.3	10	1.4	1.1	0.6	(15.6)	(13.7)	12.8	18	10.8	21.5	23 300	36 200
*60BNR10X	60	95	18	23	4.3	10	1.4	1.1	0.6	(15.6)	(13.7)	12.8	18	10.8	21.5	27 100	42 600
*60BER10E	60	95	18	–	–	–	–	1.1	0.6	15.0	13.1	22.9	25	–	26.9	16 700	23 500
*60BER10H	60	95	18	–	4.3	10	1.4	1.1	0.6	(15.0)	(13.1)	15.5	25	–	26.9	20 700	32 300
*60BER10X	60	95	18	23	4.3	10	1.4	1.1	0.6	(15.0)	(13.1)	15.5	25	–	26.9	24 600	38 800
*60BNR20EV1V	60	95	22	–	–	–	–	1.1	0.6	15.6	13.7	19.5	18	10.8	23.5	19 400	–
*60BNR20HV1V	60	95	22	–	–	–	–	1.1	0.6	(15.6)	(13.7)	12.8	18	10.8	23.5	23 300	–
*60BNR20XV1V	60	95	22	–	–	–	–	1.1	0.6	(15.6)	(13.7)	12.8	18	10.8	23.5	27 100	–
*60BER20EV1V	60	95	22	–	–	–	–	1.1	0.6	15.0	13.1	22.9	25	–	28.9	16 700	–
*60BER20HV1V	60	95	22	–	–	–	–	1.1	0.6	(15.0)	(13.1)	15.5	25	–	28.9	20 700	–
*60BER20XV1V	60	95	22	–	–	–	–	1.1	0.6	(15.0)	(13.1)	15.5	25	–	28.9	24 600	–
7212C	60	110	22	–	–	–	–	1.5	1	67.5	49.0	34.0	15	14.4	22.4	13 600	20 600
7212CSN24	60	110	22	–	–	–	–	1.5	1	(67.5)	(49.0)	40.6	15	14.4	22.4	17 700	26 900
7212A5	60	110	22	–	–	–	–	1.5	1	64.0	47.0	40.0	25	–	30.8	11 800	17 700
7212A5SN24	60	110	22	–	–	–	–	1.5	1	(64.0)	(47.0)	47.3	25	–	30.8	15 300	23 000
7212A	60	110	22	–	–	–	–	1.5	1	62.0	45.5	30.5	30	–	35.5	8 900	11 800

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다. 그리고 20, 29계열은 비접촉 씰사양만 대응 가능합니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



스핀 쇼트Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
66	79	80	1	0.5	60	113	305	581	46	60	93	128	− 1	− 5	− 15	− 25	0.195
66	79	80	1	0.5	61	122	341	661	52	68	108	149	− 1	− 5	− 15	− 25	0.175
66	79	80	1	0.5	101	201	419	884	117	150	198	267	− 2	− 5	− 10	− 18	0.198
66	79	80	1	0.5	108	223	477	1 021	133	173	231	313	− 2	− 5	− 10	− 18	0.175
66	79	80	1	0.5	49	171	343	–	47	74	97	–	0	− 8	− 16	–	0.190
66	79	80	1	0.5	49	189	388	–	53	86	113	–	0	− 8	− 16	–	0.170
66	79	80	1	0.5	49	189	388	–	53	86	113	–	0	− 8	− 16	–	0.170
66	79	80	1	0.5	49	272	557	–	78	141	183	–	0	− 8	− 15	–	0.190
66	79	80	1	0.5	49	306	638	–	88	164	215	–	0	− 8	− 15	–	0.170
66	79	80	1	0.5	49	306	638	–	88	164	215	–	0	− 8	− 15	–	0.170
66	79	80	1	0.5	49	171	343	–	47	74	97	–	0	− 8	− 16	–	0.228
66	79	80	1	0.5	49	189	388	–	53	86	113	–	0	− 8	− 16	–	0.208
66	79	80	1	0.5	49	189	388	–	53	86	113	–	0	− 8	− 16	–	0.208
66	79	80	1	0.5	49	272	557	–	78	141	183	–	0	− 8	− 15	–	0.228
66	79	80	1	0.5	49	306	638	–	88	164	215	–	0	− 8	− 15	–	0.208
66	79	80	1	0.5	49	306	638	–	88	164	215	–	0	− 8	− 15	–	0.208
67	88	90	1	0.6	96	189	526	1 092	53	70	110	157	− 4	− 10	− 25	− 42	0.405
67	88	90	1	0.6	102	208	598	1 259	60	80	128	184	− 4	− 10	− 25	− 42	0.354
67	88	90	1	0.6	162	359	780	1 549	134	179	241	317	− 4	− 9	− 17	− 28	0.408
67	88	90	1	0.6	178	407	900	1 803	155	209	283	373	− 4	− 9	− 17	− 28	0.357
67	88	90	1	0.6	49	255	929	1 371	119	206	328	379	0	− 5	− 15	− 20	0.410
67	88	90	1	0.6	49	126	348	–	51	71	105	–	0	− 5	− 15	–	0.443
67	88	90	1	0.6	49	136	393	–	57	82	123	–	0	− 5	− 15	–	0.419
67	88	90	1	0.6	49	136	393	–	57	82	123	–	0	− 5	− 15	–	0.419
67	88	90	1	0.6	49	378	775	–	85	172	224	–	0	− 10	− 18	–	0.443
67	88	90	1	0.6	49	429	894	–	95	200	263	–	0	− 10	− 18	–	0.419
67	88	90	1	0.6	49	429	894	–	95	200	263	–	0	− 10	− 18	–	0.419
67	88	90	1	0.6	49	126	348	–	51	71	105	–	0	− 5	− 15	–	0.535
67	88	90	1	0.6	49	136	393	–	57	82	123	–	0	− 5	− 15	–	0.512
67	88	90	1	0.6	49	136	393	–	57	82	123	–	0	− 5	− 15	–	0.512
67	88	90	1	0.6	49	378	775	–	85	172	224	–	0	− 10	− 18	–	0.535
67	88	90	1	0.6	49	429	894	–	95	200	263	–	0	− 10	− 18	–	0.512
67	88	90	1	0.6	49	429	894	–	95	200	263	–	0	− 10	− 18	–	0.512
69	101	104	1.5	0.8	190	397	928	1 878	64	86	126	178	− 11	− 22	− 42	− 67	0.780
69	101	104	1.5	0.8	210	449	1 069	2 183	73	100	148	209	− 11	− 22	− 42	− 67	0.644
69	101	104	1.5	0.8	293	607	1 458	2 937	155	202	282	374	− 8	− 15	− 29	− 47	0.773
69	101	104	1.5	0.8	331	698	1 697	3 441	181	237	332	441	− 8	− 15	− 29	− 47	0.637
69	101	104	1.5	0.8	49	513	1 697	2 743	112	246	380	456	0	− 10	− 25	− 35	0.789

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

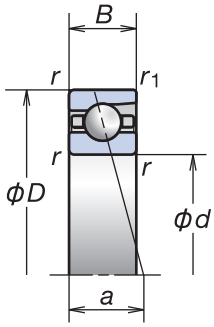
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

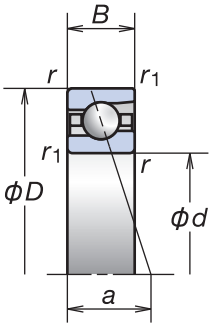
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

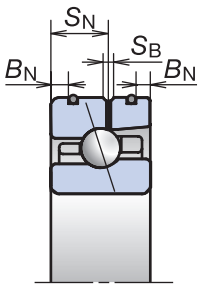
내경65mm



79,70,72계열
BNR10,20계열
BER10,20계열



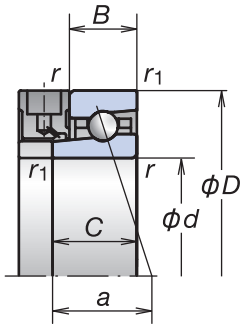
BNR19,29계열
BER19,29계열



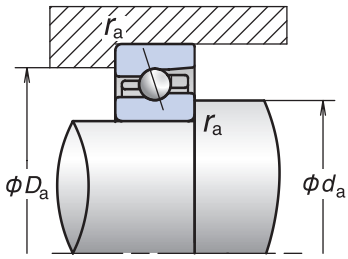
로바스트 쇼트

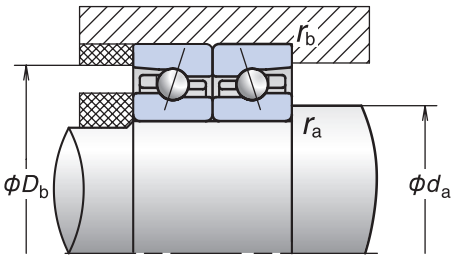
호칭 번호[1]	주요 치수[2] (mm)									기본 정격 하중[3] (kN)		허용[4] 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수[5] (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7913C	65	90	13	–	–	–	–	1	0.6	21.2	20.5	12.5	15	16.7	16.9	14 900	22 600
*7913CSN24	65	90	13	–	–	–	–	1	0.6	(21.2)	(20.5)	14.8	15	16.7	16.9	19 400	29 500
*7913A5	65	90	13	–	–	–	–	1	0.6	20.0	19.4	14.2	25	–	24.6	13 000	19 400
*7913A5SN24	65	90	13	–	–	–	–	1	0.6	(20.0)	(19.4)	16.8	25	–	24.6	16 800	25 200
*65BNR19E	65	90	13	–	–	–	–	1	0.6	15.2	13.2	18.7	18	10.8	19.1	19 400	27 700
*65BNR19H	65	90	13	–	2.8	7.5	1.4	1	0.6	(15.2)	(13.2)	12.3	18	10.8	19.1	23 300	36 200
*65BNR19X	65	90	13	18	2.8	7.5	1.4	1	0.6	(15.2)	(13.2)	12.3	18	10.8	19.1	27 100	42 600
*65BER19E	65	90	13	–	–	–	–	1	0.6	14.5	12.6	22.1	25	–	24.6	16 700	23 500
*65BER19H	65	90	13	–	2.8	7.5	1.4	1	0.6	(14.5)	(12.6)	14.9	25	–	24.6	20 700	32 300
*65BER19X	65	90	13	18	2.8	7.5	1.4	1	0.6	(14.5)	(12.6)	14.9	25	–	24.6	24 600	38 800
*65BNR29EV1V	65	90	16	–	–	–	–	1	0.6	15.2	13.2	18.7	18	10.8	20.6	19 400	–
*65BNR29HV1V	65	90	16	–	–	–	–	1	0.6	(15.2)	(13.2)	12.3	18	10.8	20.6	23 300	–
*65BNR29XV1V	65	90	16	–	–	–	–	1	0.6	(15.2)	(13.2)	12.3	18	10.8	20.6	27 100	–
*65BER29EV1V	65	90	16	–	–	–	–	1	0.6	14.5	12.6	22.1	25	–	26.1	16 700	–
*65BER29HV1V	65	90	16	–	–	–	–	1	0.6	(14.5)	(12.6)	14.9	25	–	26.1	20 700	–
*65BER29XV1V	65	90	16	–	–	–	–	1	0.6	(14.5)	(12.6)	14.9	25	–	26.1	24 600	–
*7013C	65	100	18	–	–	–	–	1.1	0.6	39.0	34.5	22.0	15	15.9	20.0	14 000	21 300
*7013CSN24	65	100	18	–	–	–	–	1.1	0.6	(39.0)	(34.5)	26.1	15	15.9	20.0	18 200	27 700
*7013A5	65	100	18	–	–	–	–	1.1	0.6	37.0	32.5	25.4	25	–	28.2	12 200	18 200
*7013A5SN24	65	100	18	–	–	–	–	1.1	0.6	(37.0)	(32.5)	30.2	25	–	28.2	15 800	23 700
*7013A	65	100	18	–	–	–	–	1.1	0.6	35.5	31.5	19.5	30	–	32.8	9 100	12 200
*65BNR10E	65	100	18	–	–	–	–	1.1	0.6	16.2	14.8	21.1	18	10.9	22.3	18 200	26 000
*65BNR10H	65	100	18	–	4.0	10.4	1.4	1.1	0.6	(16.2)	(14.8)	13.9	18	10.9	22.3	21 900	34 000
*65BNR10X	65	100	18	23	4.0	10.4	1.4	1.1	0.6	(16.2)	(14.8)	13.9	18	10.9	22.3	25 500	40 000
*65BER10E	65	100	18	–	–	–	–	1.1	0.6	15.5	14.2	24.9	25	–	28.0	15 700	22 100
*65BER10H	65	100	18	–	4.0	10.4	1.4	1.1	0.6	(15.5)	(14.2)	16.8	25	–	28.0	19 400	30 400
*65BER10X	65	100	18	23	4.0	10.4	1.4	1.1	0.6	(15.5)	(14.2)	16.8	25	–	28.0	23 100	36 400
*65BNR20EV1V	65	100	22	–	–	–	–	1.1	0.6	16.2	14.8	21.1	18	10.9	24.3	18 200	–
*65BNR20HV1V	65	100	22	–	–	–	–	1.1	0.6	(16.2)	(14.8)	13.9	18	10.9	24.3	21 900	–
*65BNR20XV1V	65	100	22	–	–	–	–	1.1	0.6	(16.2)	(14.8)	13.9	18	10.9	24.3	25 500	–
*65BER20EV1V	65	100	22	–	–	–	–	1.1	0.6	15.5	14.2	24.9	25	–	30.0	15 700	–
*65BER20HV1V	65	100	22	–	–	–	–	1.1	0.6	(15.5)	(14.2)	16.8	25	–	30.0	19 400	–
*65BER20XV1V	65	100	22	–	–	–	–	1.1	0.6	(15.5)	(14.2)	16.8	25	–	30.0	23 100	–
7213C	65	120	23	–	–	–	–	1.5	1	77.0	58.5	40.0	15	14.6	23.9	12 500	19 000
7213CSN24	65	120	23	–	–	–	–	1.5	1	(77.0)	(58.5)	47.5	15	14.6	23.9	16 300	24 700
7213A5	65	120	23	–	–	–	–	1.5	1	73.0	56.0	46.5	25	–	33.1	10 900	16 300
7213A5SN24	65	120	23	–	–	–	–	1.5	1	(73.0)	(56.0)	55.4	25	–	33.1	14 100	21 100
7213A	65	120	23	–	–	–	–	1.5	1	70.5	54.0	36.0	30	–	38.2	8 200	10 900

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다. 그리고 20, 29계열은 비접촉 씰사양만 대응 가능합니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



스핀 쇼트Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
71	84	85	1	0.5	74	151	348	690	53	71	104	145	− 2	− 7	− 16	− 27	0.208
71	84	85	1	0.5	76	164	391	787	59	81	120	169	− 2	− 7	− 16	− 27	0.186
71	84	85	1	0.5	104	212	500	948	124	161	223	288	− 2	− 5	− 11	− 18	0.211
71	84	85	1	0.5	112	235	572	1 096	142	186	261	338	− 2	− 5	− 11	− 18	0.189
71	84	85	1	0.5	49	179	364	–	50	80	105	–	0	− 8	− 16	–	0.204
71	84	85	1	0.5	49	198	412	–	56	92	122	–	0	− 8	− 16	–	0.181
71	84	85	1	0.5	49	198	412	–	56	92	122	–	0	− 8	− 16	–	0.181
71	84	85	1	0.5	49	288	595	–	83	152	198	–	0	− 8	− 15	–	0.204
71	84	85	1	0.5	49	324	683	–	93	177	232	–	0	− 8	− 15	–	0.181
71	84	85	1	0.5	49	324	683	–	93	177	232	–	0	− 8	− 15	–	0.181
71	84	85	1	0.5	49	179	364	–	50	80	105	–	0	− 8	− 16	–	0.245
71	84	85	1	0.5	49	198	412	–	56	92	122	–	0	− 8	− 16	–	0.223
71	84	85	1	0.5	49	198	412	–	56	92	122	–	0	− 8	− 16	–	0.223
71	84	85	1	0.5	49	288	595	–	83	152	198	–	0	− 8	− 15	–	0.245
71	84	85	1	0.5	49	324	683	–	93	177	232	–	0	− 8	− 15	–	0.223
71	84	85	1	0.5	49	324	683	–	93	177	232	–	0	− 8	− 15	–	0.223
72	93	95	1	0.6	130	260	537	1 062	64	85	117	163	− 6	− 13	− 24	− 39	0.435
72	93	95	1	0.6	141	290	612	1 224	73	98	136	191	− 6	− 13	− 24	− 39	0.379
72	93	95	1	0.6	209	386	915	1 781	157	196	272	356	− 5	− 9	− 18	− 29	0.455
72	93	95	1	0.6	232	439	1 059	2 077	182	229	320	420	− 5	− 9	− 18	− 29	0.399
72	93	95	1	0.6	49	272	1 012	1 498	127	225	360	418	0	− 5	− 15	− 20	0.441
72	93	95	1	0.6	49	130	367	–	54	76	113	–	0	− 5	− 15	–	0.472
72	93	95	1	0.6	49	141	415	–	60	87	131	–	0	− 5	− 15	–	0.447
72	93	95	1	0.6	49	141	415	–	60	87	131	–	0	− 5	− 15	–	0.447
72	93	95	1	0.6	49	399	824	–	89	184	240	–	0	− 10	− 18	–	0.472
72	93	95	1	0.6	49	454	952	–	100	215	282	–	0	− 10	− 18	–	0.447
72	93	95	1	0.6	49	454	952	–	100	215	282	–	0	− 10	− 18	–	0.447
72	93	95	1	0.6	49	130	367	–	54	76	113	–	0	− 5	− 15	–	0.570
72	93	95	1	0.6	49	141	415	–	60	87	131	–	0	− 5	− 15	–	0.545
72	93	95	1	0.6	49	141	415	–	60	87	131	–	0	− 5	− 15	–	0.545
72	93	95	1	0.6	49	399	824	–	89	184	240	–	0	− 10	− 18	–	0.570
72	93	95	1	0.6	49	454	952	–	100	215	282	–	0	− 10	− 18	–	0.545
72	93	95	1	0.6	49	454	952	–	100	215	282	–	0	− 10	− 18	–	0.545
74	111	114	1.5	0.8	219	448	1 069	2 175	71	95	141	200	− 12	− 23	− 44	− 70	1.01
74	111	114	1.5	0.8	243	509	1 235	2 533	82	111	165	235	− 12	− 23	− 44	− 70	0.842
74	111	114	1.5	0.8	357	657	1 664	3 307	177	221	314	414	− 9	− 15	− 30	− 48	1.00
74	111	114	1.5	0.8	405	755	1 941	3 877	207	259	370	489	− 9	− 15	− 30	− 48	0.832
74	111	114	1.5	0.8	49	553	1 851	2 998	120	269	416	499	0	− 10	− 25	− 35	1.02

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

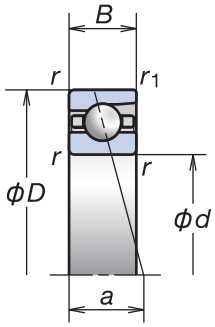
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

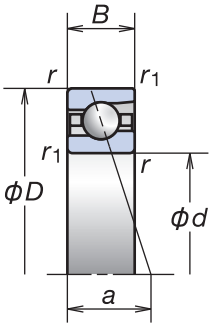
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

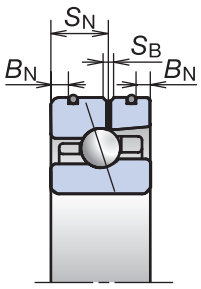
내경70mm



79,70,72계열
BNR10,20계열
BER10,20계열



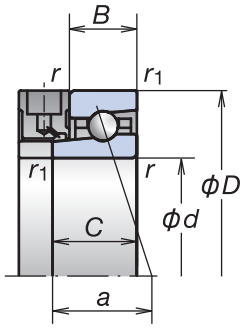
BNR19,29계열
BER19,29계열



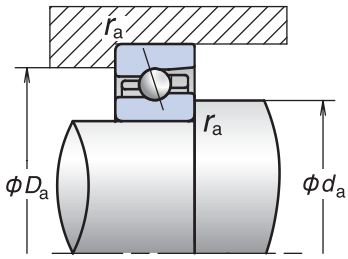
로바스트 쇼트

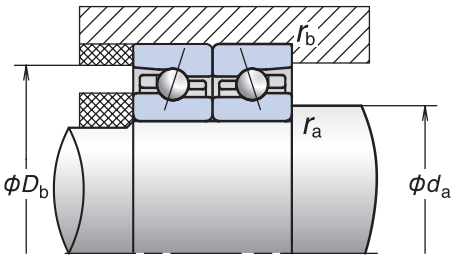
호칭 번호[1]	주요 치수[2] (mm)									기본 정격 하중[3] (kN)		허용[4] 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수[5] (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7914C	70	100	16	–	–	–	–	1	0.6	29.5	27.8	17.3	15	16.4	19.4	13 600	20 600
*7914CSN24	70	100	16	–	–	–	–	1	0.6	(29.5)	(27.8)	20.5	15	16.4	19.4	17 700	26 900
*7914A5	70	100	16	–	–	–	–	1	0.6	27.9	26.3	20.3	25	–	27.8	11 800	17 700
*7914A5SN24	70	100	16	–	–	–	–	1	0.6	(27.9)	(26.3)	24.1	25	–	27.8	15 300	23 000
*70BNR19E	70	100	16	–	–	–	–	1	0.6	21.3	18.1	26.1	18	10.8	21.8	17 700	25 200
*70BNR19H	70	100	16	–	3.1	9.3	1.4	1	0.6	(21.3)	(18.1)	17.1	18	10.8	21.8	21 200	33 000
*70BNR19X	70	100	16	21	3.1	9.3	1.4	1	0.6	(21.3)	(18.1)	17.1	18	10.8	21.8	24 800	38 900
*70BER19E	70	100	16	–	–	–	–	1	0.6	20.4	17.3	30.5	25	–	27.8	15 200	21 500
*70BER19H	70	100	16	–	3.1	9.3	1.4	1	0.6	(20.4)	(17.3)	20.7	25	–	27.8	18 900	29 500
*70BER19X	70	100	16	21	3.1	9.3	1.4	1	0.6	(20.4)	(17.3)	20.7	25	–	27.8	22 400	35 300
*70BNR29EV1V	70	100	19	–	–	–	–	1	0.6	21.3	18.1	26.1	18	10.8	23.3	17 700	–
*70BNR29HV1V	70	100	19	–	–	–	–	1	0.6	(21.3)	(18.1)	17.1	18	10.8	23.3	21 200	–
*70BNR29XV1V	70	100	19	–	–	–	–	1	0.6	(21.3)	(18.1)	17.1	18	10.8	23.3	24 800	–
*70BER29EV1V	70	100	19	–	–	–	–	1	0.6	20.4	17.3	30.5	25	–	29.3	15 200	–
*70BER29HV1V	70	100	19	–	–	–	–	1	0.6	(20.4)	(17.3)	20.7	25	–	29.3	18 900	–
*70BER29XV1V	70	100	19	–	–	–	–	1	0.6	(20.4)	(17.3)	20.7	25	–	29.3	22 400	–
*7014C	70	110	20	–	–	–	–	1.1	0.6	49.0	43.0	26.8	15	15.7	22.1	12 800	19 500
*7014CSN24	70	110	20	–	–	–	–	1.1	0.6	(49.0)	(43.0)	31.8	15	15.7	22.1	16 700	25 400
*7014A5	70	110	20	–	–	–	–	1.1	0.6	46.5	41.0	32.0	25	–	31.0	11 200	16 700
*7014A5SN24	70	110	20	–	–	–	–	1.1	0.6	(46.5)	(41.0)	38.0	25	–	31.0	14 500	21 700
*7014A	70	110	20	–	–	–	–	1.1	0.6	45.0	39.5	24.6	30	–	36.0	8 400	11 200
*70BNR10E	70	110	20	–	–	–	–	1.1	0.6	22.3	19.8	28.6	18	10.9	24.5	16 700	23 800
*70BNR10H	70	110	20	–	4.0	11.6	1.4	1.1	0.6	(22.3)	(19.8)	18.8	18	10.9	24.5	20 000	31 200
*70BNR10X	70	110	20	25	4.0	11.6	1.4	1.1	0.6	(22.3)	(19.8)	18.8	18	10.9	24.5	23 400	36 700
*70BER10E	70	110	20	–	–	–	–	1.1	0.6	21.3	18.9	33.5	25	–	30.8	14 400	20 300
*70BER10H	70	110	20	–	4.0	11.6	1.4	1.1	0.6	(21.3)	(18.9)	22.6	25	–	30.8	17 800	27 800
*70BER10X	70	110	20	25	4.0	11.6	1.4	1.1	0.6	(21.3)	(18.9)	22.6	25	–	30.8	21 200	33 400
*70BNR20EV1V	70	110	24	–	–	–	–	1.1	0.6	22.3	19.8	28.6	18	10.9	26.5	16 700	–
*70BNR20HV1V	70	110	24	–	–	–	–	1.1	0.6	(22.3)	(19.8)	18.8	18	10.9	26.5	20 000	–
*70BNR20XV1V	70	110	24	–	–	–	–	1.1	0.6	(22.3)	(19.8)	18.8	18	10.9	26.5	23 400	–
*70BER20EV1V	70	110	24	–	–	–	–	1.1	0.6	21.3	18.9	33.5	25	–	32.8	14 400	–
*70BER20HV1V	70	110	24	–	–	–	–	1.1	0.6	(21.3)	(18.9)	22.6	25	–	32.8	17 800	–
*70BER20XV1V	70	110	24	–	–	–	–	1.1	0.6	(21.3)	(18.9)	22.6	25	–	32.8	21 200	–
7214C	70	125	24	–	–	–	–	1.5	1	83.5	64.5	43.0	15	14.6	25.1	11 800	18 000
7214CSN24	70	125	24	–	–	–	–	1.5	1	(83.5)	(64.5)	51.3	15	14.6	25.1	15 400	23 400
7214A5	70	125	24	–	–	–	–	1.5	1	79.5	61.5	49.5	25	–	34.7	10 300	15 400
7214A5SN24	70	125	24	–	–	–	–	1.5	1	(79.5)	(61.5)	59.0	25	–	34.7	13 400	20 000
7214A	70	125	24	–	–	–	–	1.5	1	77.0	59.5	38.0	30	–	40.1	7 700	10 300

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다. 그리고 20, 29계열은 비접촉 씰사양만 대응 가능합니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



스핀 쇼트Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
76	94	95	1	0.5	101	205	503	1 004	59	79	119	168	− 4	− 10	− 22	− 36	0.338
76	94	95	1	0.5	108	227	571	1 155	67	91	138	196	− 4	− 10	− 22	− 36	0.301
76	94	95	1	0.5	137	298	676	1 404	138	182	248	334	− 3	− 7	− 14	− 24	0.341
76	94	95	1	0.5	150	336	777	1 633	158	212	291	392	− 3	− 7	− 14	− 24	0.304
76	94	95	1	0.5	49	181	367	–	51	80	105	–	0	− 8	− 16	–	0.328
76	94	95	1	0.5	49	200	415	–	57	93	122	–	0	− 8	− 16	–	0.292
76	94	95	1	0.5	49	200	415	–	57	93	122	–	0	− 8	− 16	–	0.292
76	94	95	1	0.5	49	292	604	–	84	154	201	–	0	− 8	− 15	–	0.328
76	94	95	1	0.5	49	329	693	–	94	180	235	–	0	− 8	− 15	–	0.292
76	94	95	1	0.5	49	329	693	–	94	180	235	–	0	− 8	− 15	–	0.292
76	94	95	1	0.5	49	181	367	–	51	80	105	–	0	− 8	− 16	–	0.381
76	94	95	1	0.5	49	200	415	–	57	93	122	–	0	− 8	− 16	–	0.344
76	94	95	1	0.5	49	200	415	–	57	93	122	–	0	− 8	− 16	–	0.344
76	94	95	1	0.5	49	292	604	–	84	154	201	–	0	− 8	− 15	–	0.381
76	94	95	1	0.5	49	329	693	–	94	180	235	–	0	− 8	− 15	–	0.344
76	94	95	1	0.5	49	329	693	–	94	180	235	–	0	− 8	− 15	–	0.344
77	103	105	1	0.6	148	285	732	1 460	68	89	135	190	− 7	− 14	− 30	− 48	0.606
77	103	105	1	0.6	161	319	839	1 692	78	103	158	222	− 7	− 14	− 30	− 48	0.525
77	103	105	1	0.6	255	500	1 080	2 196	172	220	294	391	− 6	− 11	− 20	− 33	0.625
77	103	105	1	0.6	285	572	1 252	2 566	200	257	346	461	− 6	− 11	− 20	− 33	0.544
77	103	105	1	0.6	49	278	1 038	2 106	130	231	370	482	0	− 5	− 15	− 25	0.613
77	103	105	1	0.6	49	235	509	–	53	93	126	–	0	− 10	− 20	–	0.645
77	103	105	1	0.6	49	262	582	–	60	108	147	–	0	− 10	− 20	–	0.605
77	103	105	1	0.6	49	262	582	–	60	108	147	–	0	− 10	− 20	–	0.605
77	103	105	1	0.6	49	396	815	–	89	181	236	–	0	− 10	− 18	–	0.645
77	103	105	1	0.6	49	451	942	–	100	212	278	–	0	− 10	− 18	–	0.605
77	103	105	1	0.6	49	451	942	–	100	212	278	–	0	− 10	− 18	–	0.605
77	103	105	1	0.6	49	235	509	–	53	93	126	–	0	− 10	− 20	–	0.764
77	103	105	1	0.6	49	262	582	–	60	108	147	–	0	− 10	− 20	–	0.724
77	103	105	1	0.6	49	262	582	–	60	108	147	–	0	− 10	− 20	–	0.724
77	103	105	1	0.6	49	396	815	–	89	181	236	–	0	− 10	− 18	–	0.764
77	103	105	1	0.6	49	451	942	–	100	212	278	–	0	− 10	− 18	–	0.724
77	103	105	1	0.6	49	451	942	–	100	212	278	–	0	− 10	− 18	–	0.724
79	116	119	1.5	0.8	243	484	1 164	2 368	75	100	148	209	− 9	− 20	− 42	− 69	1.09
79	116	119	1.5	0.8	262	539	1 329	2 734	86	115	172	245	− 9	− 20	− 42	− 69	0.898
79	116	119	1.5	0.8	406	781	1 856	3 690	188	239	332	438	− 8	− 15	− 30	− 49	1.08
79	116	119	1.5	0.8	451	887	2 147	4 304	218	279	390	517	− 8	− 15	− 30	− 49	0.888
79	116	119	1.5	0.8	98	1 067	2 626	3 932	153	346	481	562	0	− 15	− 30	− 40	1.10

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

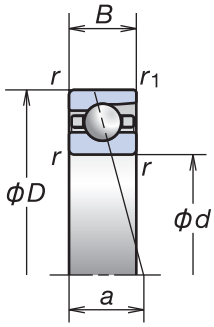
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

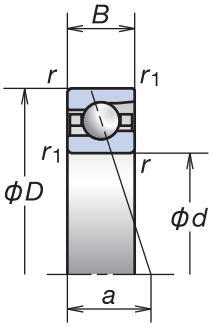
- 동등가 하중 ····· P191
- 정등가 하중 ····· P198
- 스페이서 · 노즐 목표 위치 ····· P237
- 그리스 추천 봉입량 ····· P257

1. 앵귤러 볼 베어링

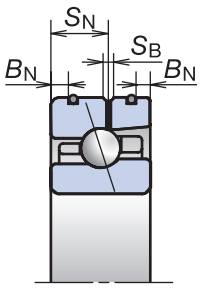
내경75mm



79,70,72계열
BNR10,20계열
BER10,20계열



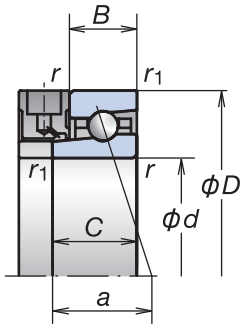
BNR19,29계열
BER19,29계열



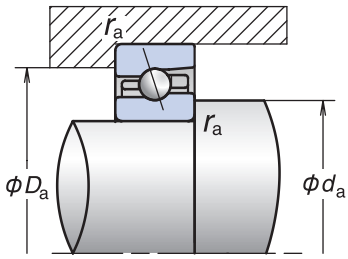
로바스트 쇼트

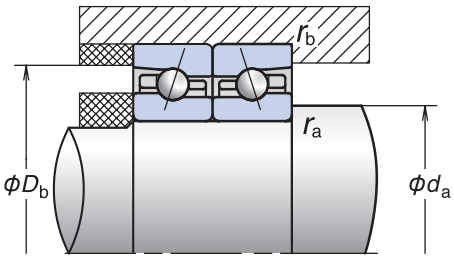
호칭 번호[1]	주요 치수[2] (mm)									기본 정격 하중[3] (kN)		허용[4] 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수[5] (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7915C	75	105	16	–	–	–	–	1	0.6	30.0	29.3	18.0	15	16.6	20.1	12 800	19 500
*7915CSN24	75	105	16	–	–	–	–	1	0.6	(30.0)	(29.3)	21.3	15	16.6	20.1	16 700	25 400
*7915A5	75	105	16	–	–	–	–	1	0.6	28.3	27.7	21.2	25	–	29.0	11 200	16 700
*7915A5SN24	75	105	16	–	–	–	–	1	0.6	(28.3)	(27.7)	25.1	25	–	29.0	14 500	21 700
*75BNR19E	75	105	16	–	–	–	–	1	0.6	21.6	19.0	27.5	18	10.9	22.6	16 700	23 800
*75BNR19H	75	105	16	–	3.1	9.3	1.4	1	0.6	(21.6)	(19.0)	18.0	18	10.9	22.6	20 000	31 200
*75BNR19X	75	105	16	21	3.1	9.3	1.4	1	0.6	(21.6)	(19.0)	18.0	18	10.9	22.6	23 400	36 700
*75BER19E	75	105	16	–	–	–	–	1	0.6	20.7	18.2	32.5	25	–	29.0	14 400	20 300
*75BER19H	75	105	16	–	3.1	9.3	1.4	1	0.6	(20.7)	(18.2)	21.7	25	–	29.0	17 800	27 800
*75BER19X	75	105	16	21	3.1	9.3	1.4	1	0.6	(20.7)	(18.2)	21.7	25	–	29.0	21 200	33 400
*75BNR29EV1V	75	105	19	–	–	–	–	1	0.6	21.6	19.0	27.5	18	10.9	24.1	16 700	–
*75BNR29HV1V	75	105	19	–	–	–	–	1	0.6	(21.6)	(19.0)	18.0	18	10.9	24.1	20 000	–
*75BNR29XV1V	75	105	19	–	–	–	–	1	0.6	(21.6)	(19.0)	18.0	18	10.9	24.1	23 400	–
*75BER29EV1V	75	105	19	–	–	–	–	1	0.6	20.7	18.2	32.5	25	–	30.5	14 400	–
*75BER29HV1V	75	105	19	–	–	–	–	1	0.6	(20.7)	(18.2)	21.7	25	–	30.5	17 800	–
*75BER29XV1V	75	105	19	–	–	–	–	1	0.6	(20.7)	(18.2)	21.7	25	–	30.5	21 200	–
*7015C	75	115	20	–	–	–	–	1.1	0.6	50.5	45.5	28.1	15	15.9	22.7	12 200	18 500
*7015CSN24	75	115	20	–	–	–	–	1.1	0.6	(50.5)	(45.5)	33.3	15	15.9	22.7	15 800	24 000
*7015A5	75	115	20	–	–	–	–	1.1	0.6	47.5	43.5	33.5	25	–	32.1	10 600	15 800
*7015A5SN24	75	115	20	–	–	–	–	1.1	0.6	(47.5)	(43.5)	40.0	25	–	32.1	13 700	20 600
*7015A	75	115	20	–	–	–	–	1.1	0.6	46.0	41.5	25.9	30	–	37.4	7 900	10 600
*75BNR10E	75	115	20	–	–	–	–	1.1	0.6	22.6	20.7	30.0	18	11.0	25.3	15 800	22 600
*75BNR10H	75	115	20	–	4.0	11.6	1.4	1.1	0.6	(22.6)	(20.7)	19.7	18	11.0	25.3	19 000	29 500
*75BNR10X	75	115	20	27	4.0	11.6	1.4	1.1	0.6	(22.6)	(20.7)	19.7	18	11.0	25.3	22 200	34 800
*75BER10E	75	115	20	–	–	–	–	1.1	0.6	21.6	19.8	35.0	25	–	31.9	13 600	19 200
*75BER10H	75	115	20	–	4.0	11.6	1.4	1.1	0.6	(21.6)	(19.8)	23.7	25	–	31.9	16 900	26 400
*75BER10X	75	115	20	27	4.0	11.6	1.4	1.1	0.6	(21.6)	(19.8)	23.7	25	–	31.9	20 000	31 600
*75BNR20EV1V	75	115	24	–	–	–	–	1.1	0.6	22.6	20.7	30.0	18	11.0	27.3	15 800	–
*75BNR20HV1V	75	115	24	–	–	–	–	1.1	0.6	(22.6)	(20.7)	19.7	18	11.0	27.3	19 000	–
*75BNR20XV1V	75	115	24	–	–	–	–	1.1	0.6	(22.6)	(20.7)	19.7	18	11.0	27.3	22 200	–
*75BER20EV1V	75	115	24	–	–	–	–	1.1	0.6	21.6	19.8	35.0	25	–	33.9	13 600	–
*75BER20HV1V	75	115	24	–	–	–	–	1.1	0.6	(21.6)	(19.8)	23.7	25	–	33.9	16 900	–
*75BER20XV1V	75	115	24	–	–	–	–	1.1	0.6	(21.6)	(19.8)	23.7	25	–	33.9	20 000	–
7215C	75	130	25	–	–	–	–	1.5	1	87.0	70.0	46.0	15	14.8	26.2	11 300	17 100
7215CSN24	75	130	25	–	–	–	–	1.5	1	(87.0)	(70.0)	54.9	15	14.8	26.2	14 700	22 300
7215A5	75	130	25	–	–	–	–	1.5	1	82.5	66.5	53.0	25	–	36.4	9 800	14 700
7215A5SN24	75	130	25	–	–	–	–	1.5	1	(82.5)	(66.5)	62.7	25	–	36.4	12 700	19 100
7215A	75	130	25	–	–	–	–	1.5	1	80.0	64.5	40.5	30	–	42.1	7 400	9 800

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다. 그리고 20, 29계열은 비접촉 씰사양만 대응 가능합니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.



스핀 쇼트Ⅱ





설치 관계 치수 (mm)					예압 하중(DB, DF조합) (N)				액셜 강성(DB, DF조합) (N/μm)				측정 액셜 클리어런스 (μm)				질량 (kg) (참고)
d_a (최소)	D_a (최대)	D_b (최대)	r_a (최대)	r_b (최대)	EL	L	M	H	EL	L	M	H	EL	L	M	H	
81	99	100	1	0.5	103	190	489	997	61	79	120	171	− 4	− 9	− 21	− 35	0.358
81	99	100	1	0.5	109	210	555	1 147	70	91	140	199	− 4	− 9	− 21	− 35	0.319
81	99	100	1	0.5	140	307	765	1 458	142	189	268	347	− 3	− 7	− 15	− 24	0.355
81	99	100	1	0.5	153	346	882	1 696	164	220	314	409	− 3	− 7	− 15	− 24	0.316
81	99	100	1	0.5	49	185	464	–	52	83	118	–	0	− 8	− 19	–	0.348
81	99	100	1	0.5	49	205	528	–	58	96	138	–	0	− 8	− 19	–	0.310
81	99	100	1	0.5	49	205	528	–	58	96	138	–	0	− 8	− 19	–	0.310
81	99	100	1	0.5	49	301	625	–	87	160	209	–	0	− 8	− 15	–	0.348
81	99	100	1	0.5	49	339	718	–	97	187	245	–	0	− 8	− 15	–	0.310
81	99	100	1	0.5	49	339	718	–	97	187	245	–	0	− 8	− 15	–	0.310
81	99	100	1	0.5	49	185	464	–	52	83	118	–	0	− 8	− 19	–	0.403
81	99	100	1	0.5	49	205	528	–	58	96	138	–	0	− 8	− 19	–	0.365
81	99	100	1	0.5	49	205	528	–	58	96	138	–	0	− 8	− 19	–	0.365
81	99	100	1	0.5	49	301	625	–	87	160	209	–	0	− 8	− 15	–	0.403
81	99	100	1	0.5	49	339	718	–	97	187	245	–	0	− 8	− 15	–	0.365
81	99	100	1	0.5	49	339	718	–	97	187	245	–	0	− 8	− 15	–	0.365
82	108	110	1	0.6	151	294	796	1 573	70	92	144	202	− 7	− 14	− 31	− 49	0.643
82	108	110	1	0.6	165	330	915	1 824	81	107	168	236	− 7	− 14	− 31	− 49	0.557
82	108	110	1	0.6	263	519	1 204	2 399	179	230	316	418	− 6	− 11	− 21	− 34	0.652
82	108	110	1	0.6	295	594	1 398	2 806	209	269	372	493	− 6	− 11	− 21	− 34	0.566
82	108	110	1	0.6	49	288	1 083	2 202	135	241	387	506	0	− 5	− 15	− 25	0.650
82	108	110	1	0.6	49	240	525	–	55	96	131	–	0	− 10	− 20	–	0.679
82	108	110	1	0.6	49	269	600	–	61	112	153	–	0	− 10	− 20	–	0.638
82	108	110	1	0.6	49	269	600	–	61	112	153	–	0	− 10	− 20	–	0.638
82	108	110	1	0.6	49	506	1 034	–	91	203	265	–	0	− 12	− 21	–	0.679
82	108	110	1	0.6	49	579	1 199	–	103	238	311	–	0	− 12	− 21	–	0.638
82	108	110	1	0.6	49	579	1 199	–	103	238	311	–	0	− 12	− 21	–	0.638
82	108	110	1	0.6	49	240	525	–	55	96	131	–	0	− 10	− 20	–	0.806
82	108	110	1	0.6	49	269	600	–	61	112	153	–	0	− 10	− 20	–	0.764
82	108	110	1	0.6	49	269	600	–	61	112	153	–	0	− 10	− 20	–	0.764
82	108	110	1	0.6	49	506	1 034	–	91	203	265	–	0	− 12	− 21	–	0.806
82	108	110	1	0.6	49	579	1 199	–	103	238	311	–	0	− 12	− 21	–	0.764
82	108	110	1	0.6	49	579	1 199	–	103	238	311	–	0	− 12	− 21	–	0.764
84	121	124	1.5	0.8	270	530	1 224	2 445	81	108	157	220	− 10	− 21	− 42	− 68	1.19
84	121	124	1.5	0.8	293	593	1 398	2 825	93	124	182	257	− 10	− 21	− 42	− 68	0.983
84	121	124	1.5	0.8	422	819	1 961	3 911	199	253	353	467	− 8	− 15	− 30	− 49	1.18
84	121	124	1.5	0.8	469	932	2 270	4 565	231	296	415	550	− 8	− 15	− 30	− 49	0.973
84	121	124	1.5	0.8	98	1 123	2 780	4 170	159	367	512	598	0	− 15	− 30	− 40	1.20

레이디얼 강성 산출
액셜 강성 값에 A표의 계수를 곱하여 산출합니다.

다열 조합의 산출
예압 하중, 액셜 강성 값은 각각 B표의 계수를 곱하여 산출합니다.
레이디얼 강성 값은 A표에서 구한 값에 B표의 계수를 곱하여 산출합니다.

A표

	EL	L	M	H
15°	6.5	6.0	5.0	4.5
18°	4.5			
25°	2.0			
30°	1.4			

B표

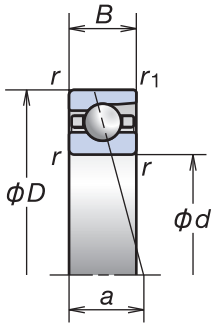
	DBD	DBB
예압 계수	1.36	2
액셜 강성	1.48	2
레이디얼 강성	1.54	2

관련 자료 참조 페이지

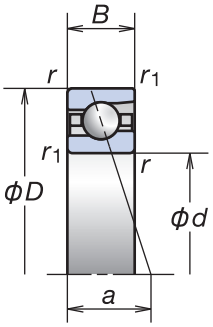
- 동등가 하중 …… P191
- 정등가 하중 …… P198
- 스페이서 · 노즐 목표 위치 …… P237
- 그리스 추천 봉입량 …… P257

1. 앵귤러 볼 베어링

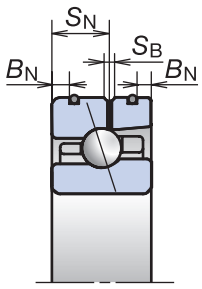
내경80mm



79,70,72계열
BNR10,20계열
BER10,20계열



BNR19,29계열
BER19,29계열



로바스트 쇼트

호칭 번호[1]	주요 치수[2] (mm)									기본 정격 하중[3] (kN)		허용[4] 액셜 하중 (kN)	접촉각 (도)	계수 f_o	작용점 위치 (mm) a	허용 회전수[5] (min^{-1})	
	d	D	B	C	B_N	S_N	S_B	r (최소)	r_1 (최소)	C_r (동정격)	C_{or} (정정격)					그리스 윤활	오일 윤활
*7916C	80	110	16	–	–	–	–	1	0.6	30.5	30.5	18.7	15	16.7	20.7	12 200	18 500
*7916CSN24	80	110	16	–	–	–	–	1	0.6	(30.5)	(30.5)	22.2	15	16.7	20.7	15 800	24 000
*7916A5	80	110	16	–	–	–	–	1	0.6	28.7	29.0	22.1	25	–	30.2	10 600	15 800
*7916A5SN24	80	110	16	–	–	–	–	1	0.6	(28.7)	(29.0)	26.2	25	–	30.2	13 700	20 600
*80BNR19E	80	110	16	–	–	–	–	1	0.6	22.0	19.9	28.9	18	11.0	23.4	15 800	22 600
*80BNR19H	80	110	16	–	3.1	9.3	1.4	1	0.6	(22.0)	(19.9)	18.9	18	11.0	23.4	19 000	29 500
*80BNR19X	80	110	16	21	3.1	9.3	1.4	1	0.6	(22.0)	(19.9)	18.9	18	11.0	23.4	22 200	34 800
*80BER19E	80	110	16	–	–	–	–	1	0.6	21.0	19.1	34.0	25	–	30.2	13 600	19 200
*80BER19H	80	110	16	–	3.1	9.3	1.4	1	0.6	(21.0)	(19.1)	22.8	25	–	30.2	16 900	26 400
*80BER19X	80	110	16	21	3.1	9.3	1.4	1	0.6	(21.0)	(19.1)	22.8	25	–	30.2	20 000	31 600
*80BNR29EV1V	80	110	19	–	–	–	–	1	0.6	22.0	19.9	28.9	18	11.0	24.9	15 800	–
*80BNR29HV1V	80	110	19	–	–	–	–	1	0.6	(22.0)	(19.9)	18.9	18	11.0	24.9	19 000	–
*80BNR29XV1V	80	110	19	–	–	–	–	1	0.6	(22.0)	(19.9)	18.9	18	11.0	24.9	22 200	–
*80BER29EV1V	80	110	19	–	–	–	–	1	0.6	21.0	19.1	34.0	25	–	31.7	13 600	–
*80BER29HV1V	80	110	19	–	–	–	–	1	0.6	(21.0)	(19.1)	22.8	25	–	31.7	16 900	–
*80BER29XV1V	80	110	19	–	–	–	–	1	0.6	(21.0)	(19.1)	22.8	25	–	31.7	20 000	–
*7016C	80	125	22	–	–	–	–	1.1	0.6	61.5	55.5	34.5	15	15.7	24.7	11 300	17 100
*7016CSN24	80	125	22	–	–	–	–	1.1	0.6	(61.5)	(55.5)	40.9	15	15.7	24.7	14 700	22 300
*7016A5	80	125	22	–	–	–	–	1.1	0.6	58.5	52.5	41.0	25	–	34.9	9 800	14 700
*7016A5SN24	80	125	22	–	–	–	–	1.1	0.6	(58.5)	(52.5)	48.3	25	–	34.9	12 700	19 100
*7016A	80	125	22	–	–	–	–	1.1	0.6	56.0	50.5	31.5	30	–	40.6	7 400	9 800
*80BNR10E	80	125	22	–	–	–	–	1.1	0.6	26.5	24.5	35.5	18	10.9	27.5	14 700	20 900
*80BNR10H	80	125	22	–	4.7	12.2	2.2	1.1	0.6	(26.5)	(24.5)	23.4	18	10.9	27.5	17 600	27 400
*80BNR10X	80	125	22	27	4.7	12.2	2.2	1.1	0.6	(26.5)	(24.5)	23.4	18	10.9	27.5	20 500	32 200
*80BER10E	80	125	22	–	–	–	–	1.1	0.6	25.3	23.5	42.0	25	–	34.6	12 600	17 800
*80BER10H	80	125	22	–	4.7	12.2	2.2	1.1	0.6	(25.3)	(23.5)	28.2	25	–	34.6	15 700	24 400
*80BER10X	80	125	22	27	4.7	12.2	2.2	1.1	0.6	(25.3)	(23.5)	28.2	25	–	34.6	18 600	29 300
*80BNR20EV1V	80	125	27	–	–	–	–	1.1	0.6	26.5	24.5	35.5	18	10.9	30.0	14 700	–
*80BNR20HV1V	80	125	27	–	–	–	–	1.1	0.6	(26.5)	(24.5)	23.4	18	10.9	30.0	17 600	–
*80BNR20XV1V	80	125	27	–	–	–	–	1.1	0.6	(26.5)	(24.5)	23.4	18	10.9	30.0	20 500	–
*80BER20EV1V	80	125	27	–	–	–	–	1.1	0.6	25.3	23.5	42.0	25	–	37.1	12 600	–
*80BER20HV1V	80	125	27	–	–	–	–	1.1	0.6	(25.3)	(23.5)	28.2	25	–	37.1	15 700	–
*80BER20XV1V	80	125	27	–	–	–	–	1.1	0.6	(25.3)	(23.5)	28.2	25	–	37.1	18 600	–
7216C	80	140	26	–	–	–	–	2	1	97.5	77.5	54.5	15	14.7	27.7	10 500	16 000
7216CSN24	80	140	26	–	–	–	–	2	1	(97.5)	(77.5)	64.5	15	14.7	27.7	13 700	20 800
7216A5	80	140	26	–	–	–	–	2	1	93.0	74.0	62.0	25	–	38.6	9 100	13 700
7216A5SN24	80	140	26	–	–	–	–	2	1	(93.0)	(74.0)	73.5	25	–	38.6	11 900	17 800
7216A	80	140	26	–	–	–	–	2	1	89.5	71.5	47.5	30	–	44.8	6 900	9 100

주 (1)*표시가 있는 베어링은 비접촉 씰사양 대응품입니다. 그리고 20, 29계열은 비접촉 씰사양만 대응 가능합니다.
(2) C값이 기재된 것은 스핀 쇼트Ⅱ, 그리고 BN, SN, SB의 값이 기재된 것은 로바스트 쇼트 대응품입니다.
(3) 세라믹 볼 베어링의 정격 하중은 참고치입니다.
(4) 허용 액셜 하중에 대해서는 P199를 참조하십시오.
(5) 허용 회전수의 적용에 대해서는 P216을 참조하십시오.